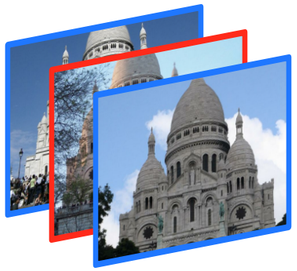
The CAPRI computational pipeline

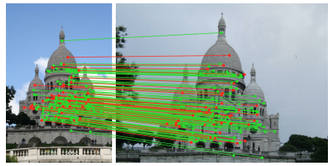
A certification layer between sparse and dense reconstruction. It replaces no stage.

1 Images



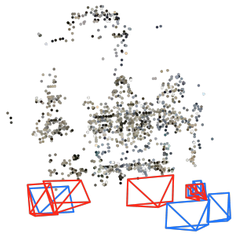
Uncalibrated photos, split into view subsets V_A and V_B

2 Matches



SIFT or SuperGlue features; fundamental matrix (MAGSAC)

3 Cameras and points



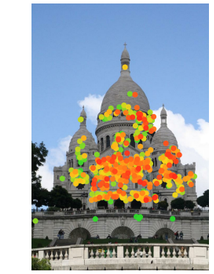
Projective SfM, bundle adjustment, Tyler frame

4 CAPRI layer

1. Conditioning gate
2. Tuple selection
3. Strata
4. Consistency test

Milliseconds per scene; details below

5 Certified points



low high pencil score

Median score: 0.83 on jittered points, 0.72 on the others; plus abstentions and S

6 Dense MVS



Unchanged, run on the certified points

Inside the CAPRI layer

1 Conditioning gate

$\kappa_J = \max_k \|\nabla \log|J_k|\|$ from the rows of M^{-1} (Jacobi's formula), 15 inverses per tuple.
Keep tuples with $\kappa_J \le \kappa_{\max}$; singular brackets ($< 10^{-12}$) get $\kappa_J = \infty$.

10,000 tuples: 3.8 ms (C++), 132 ms (NumPy), 2.0 s (Python loop)

2 Tuple selection

Greedy maximum coverage over κ-feasible tuples. Coverage is monotone submodular, so greedy reaches $\ge (1 - 1/e)\ \mathrm{OPT}$.
Batched gains from a boolean membership matrix; ties go to the smaller κ_J.

$n = 30$, 10,000 candidates: 38.5 ms (batched) vs 2.1 s (per tuple)

3 Strata

Per selected tuple: the size of its largest coplanar subset (rank classifier). The coplanar strata have codimension $m - 3$.

Critical tuple ⇒ rejected.
The consistency test abstains on (near) planar point sets.

4 Consistency test

Reconstruct V_A and V_B separately. The $(n \text{ choose } 4)$ log-dets give every $\log|\mathrm{CR}_t|$ by table lookup ($n = 12$: $495 \to 41{,}580$).
For $n > 12$: 50,000 uniform pencils; the sampled $Q_{0.1}$ lies in $[Q_{0.09},\ Q_{0.11}]$ w.p. $\ge 1 - 10^{-4}$ (DKW).

1.5 ms ($n = 12$), 15.0 ms ($n = 300$), C++

Implementation: Python reference and C++ kernels (pybind11, Eigen fixed-size 4×4, GIL released) with identical outputs.
Every result file records its seed, backend and commit. Runtimes: one CPU core, median of five runs. Matching and bundle adjustment dominate the cost.